

Calculation of the field distribution of a Ti:LiNbO₃ optical waveguide and its applications

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A trial solution for the dominant field distribution of a Ti:LiNbO₃ optical waveguide is proposed in this paper, and its parameters are determined by using the variational method. Comparison with the effective index method shows that this solution is a good approximation for the fundamental and the higher-order modes. The approximate solution of the fundamental mode is employed to calculate the power coupling coefficient of the step-profile fibre and optical waveguide. For the waveguide with typical diffusion parameters, the maximum coupling efficiency is found to be 80%. The interaction or overlap integral between the optical mode and the applied signal is investigated. The two optical waveguides of a Z-cut amplitude modulator should be placed 1.1 and 1.7 μm apart from the assumed appropriate positions.

1. Introduction

In recent years waveguide electrooptic modulators have drawn more and more attention, and significant effort has been made to achieve high fibre-waveguide-fibre coupling efficiency and low required drive voltage. The calculation of the guided modes is the basis of analysing and optimizing the integrated optical devices. The analytical expression of the field components is especially attractive because it is convenient and much faster than the numerical finite-element method. Marcuse [1] approximated the x and y dependence of the transverse field component of the fundamental mode by Gaussian and Hermite-Gaussian functions, respectively. The mode sizes were calculated directly from controllable fabrication parameters by employing a variational method [2]. Here we give a new trial solution which can be applied to the description of the fundamental and the higher-order guided modes. To assess the quality of this approximation, we compare the propagation constants obtained by the variational method with those of the effective index method [3]. The excellent agreement shows that the analytical solution gives a good approximation.

Several methods of calculating fibre-waveguide insertion loss have been reported [4–6]. Because the waveguide depth mode is quite asymmetric, the assumption that the waveguide mode is modelled as a simple Gaussian [6] will lead to error. Ramer [7] calculated the change

in the optical propagation constant for a variety of electrode configurations. In this analysis, the waveguide modes approximated by Marcatili [8] were used. Owing to the rough approximations of the waveguide modes, the analysis results can only be used as reference. Marcuse [1] optimized the spacing of the electrodes and the positions for a single waveguide placed beneath semi-infinite electrodes. In practice, the electrode configurations widely used for integrated optical devices are coplanar waveguide and coplanar strips.

In this paper we derive the power coupling coefficient from the approximate solution for waveguide mode and the exact expressions for guided modes of an optical fibre with infinite cladding. The maximum coupling coefficient and the optimum location of the fibre are found from the formula developed. In addition, the operation between the applied electric field and optical mode is investigated in order to optimize the optical waveguide positions and provide design data for choosing the diffusion parameters.

2. Approximation of the field distribution

For a Ti: LiNbO₃ channel waveguide with W the initial width of the Ti strip, the refractive index profile can be expressed as [3]:

$$n^2(x, y) = \begin{cases} 1 & y < 0 \\ n_b^2 + (n_s^2 - n_b^2)f(y/D_y)g(2x/W) & y > 0 \end{cases} \quad (1a)$$

where

$$f(y/D_y) = \exp(-y^2/D_y^2) \quad (1b)$$

$$g(2x/W) = \frac{1}{2} \{ \operatorname{erf}[W/2D_x(1 + 2x/W)] + \operatorname{erf}[W/2D_x(1 - 2x/W)] \} \quad (1c)$$

n_b is the refractive index of the substrate, n_s is the refractive index at $x = y = 0$ for a sufficiently wide Ti strip. D_x and D_y are the diffusion width and depth, respectively. Based on Marcuse's assumption [1], we express the predominant electric field component as

$$\phi = \frac{2}{(W_x W_y)^{1/2}} \psi_n(2^{1/2}x/W_x) \psi_{2m+1}(\sqrt{2}^{1/2}y/W_y) \quad (2a)$$

where

$$\psi_n(\xi) = \frac{1}{(2^n n! \sqrt{\pi})^{1/2}} \exp(-\xi^2/2) H_n(\xi) \quad (2b)$$

$H_n(\xi)$ is the Hermite polynomial of the n th order; n and $2m + 1$ are the mode orders in the x and y directions, respectively. Usually, there exists only the lowest-order guided mode in the depth direction, as applies to most cases. Thus we only consider the case $m = 0$. W_x and W_y are the width and depth parameters, which can be determined by the variational method [9]. Generally, W_x and W_y are no longer the mode $1/e$ intensity width and depth. If ϕ is a good approximation to the mode profile, it can be used to find the propagation constant β by

$$\beta^2 = \frac{\iint \phi \nabla_t^2 \phi \, dx \, dy + k^2 \iint n^2 \phi^2 \, dx \, dy}{\iint \phi^2 \, dx \, dy} \quad (3)$$

The optimum values of W_x and W_y correspond to the largest value of β^2 . Substituting Equation 2 into Equation 3, we have

$$\beta^2 = k^2 n_b^2 + \frac{2^{3/2}(n_s^2 - n_b^2)k^2}{W_x} \left(1 + \frac{W_y^2}{2D_y^2} \right)^{-3/2} Q - \frac{2n + 1}{W_x^2} - \frac{3}{W_y^2} \quad (4a)$$

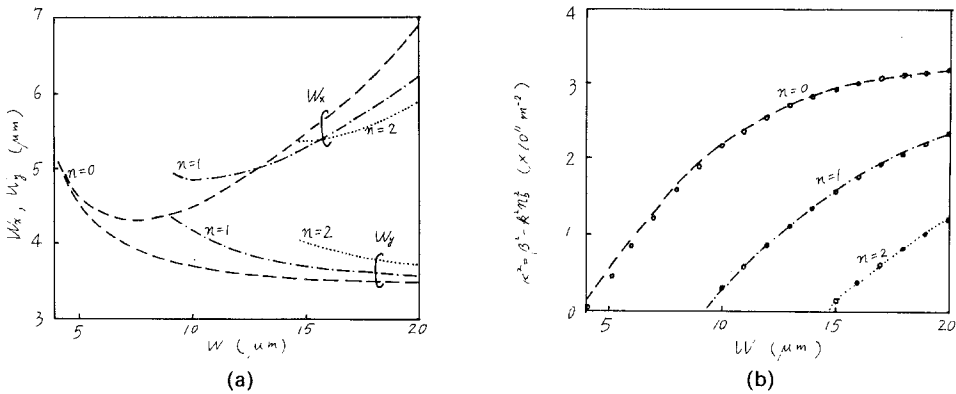


Figure 1 (a) Mode sizes calculated by the variational method versus Ti strip width; (b) the corresponding propagation constants, compared with those of the effective index method (open circle).

where

$$Q = \int_0^\infty g(2x/W) \psi_n^2(2^{1/2}x/W_x) dx \quad (4b)$$

In the case $m = n = 0$, the above variational equation is equivalent to that reported in [2], and Equation 2 reduces to Equation 5 in [1], noting that we still follow the former definitions of parameters W_x and W_y [1]. Figure 1 gives the parameters and the corresponding propagation constants as functions of Ti strip width. The results of the effective index method are also plotted in Fig. 1b. From Fig. 1b we see that the waveguide with a 4 to 9 μm wide Ti strip supports only the fundamental modes. Considering that β^2 is much larger than $\beta^2 - k^2 n_b^2$, we may conclude that Equation 2 gives a good approximation for the fundamental and the higher-order lateral modes.

3. Fibre-waveguide coupling

End-butt coupling of single-mode fibre to single-mode waveguide is particularly useful because of its convenience and high efficiency. In this section we utilize the approximate solution discussed in Section 2 to establish the relation between the coupling efficiency and the diffusion parameters. For the case of a step-profile fibre, operating in the y-polarized HE₁₁ mode and coupled to a weakly guiding channel waveguide, as shown in Fig. 2, the

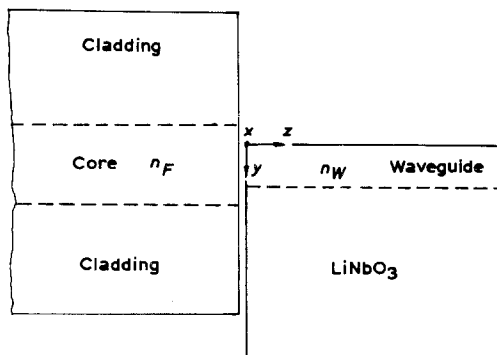


Figure 2 End-butt coupling of a step-profile fibre and optical waveguide.

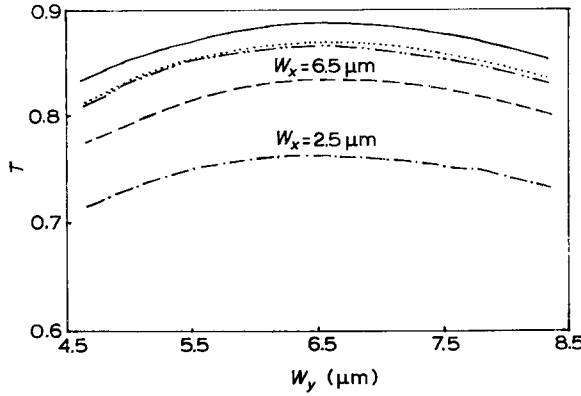


Figure 3 Calculated coupling coefficient as a function of mode parameters. —, $W_y = 4.5 \mu\text{m}$; $\cdots$, $W_y = 5.5 \mu\text{m}$; — · — · —, $W_y = 3.5 \mu\text{m}$.

power coupling coefficient for the lowest-order TM mode is easily obtained:

$$T = \frac{4n_f n_w}{(n_f + n_w)^2} \left| \iint \phi E_y dx dy \right|^2 \quad (5)$$

The first factor arises from reflection at the fibre-waveguide interface. It could be reduced by using an antireflection coating on end faces or index matching fluid between the fibre and waveguide. In the analysis below we disregard the effect of this term. The overlap integral accounts for mismatch in the field distributions in the fibre and waveguide. E_y represents the transverse field of the fibre and is given by the well-known equation

$$E_y = \begin{cases} AJ_0(\chi r) & r \leq a \\ AJ_0(\chi a)K_0(vr)/K_0(va) & r \geq a \end{cases} \quad (6)$$

The normalization coefficient A is

$$A = \{v^2/[a^2\pi NA^2 k J_1^2(\chi a)]\}^{1/2} \quad (7)$$

Typically, for a single-mode fibre at $\lambda = 1.32 \mu\text{m}$, the radius of the core is $4 \mu\text{m}$, and the numerical aperture NA is 0.12. Calculated coupling coefficient as a function of mode parameters is plotted in Fig. 3. The optimum power coupling efficiency can approach 87% for $W_x = 4.5 \mu\text{m}$, and $W_y = 6.7 \mu\text{m}$. The expected mode sizes cannot be realized by adjusting the diffusion parameters according to the analysis of Section 2, but the results indicate that relatively deep diffusion is required to achieve higher coupling efficiency. For a Z-cut LiNbO₃ optical waveguide having the following parameters: $D_x = 3.6 \mu\text{m}$, $D_y = 4.0 \mu\text{m}$ [10], index change $\Delta n = n_s - n_b = 0.01$, and $W = 6 \mu\text{m}$, the mode parameters W_x and W_y are found to be 4.4 and $4.2 \mu\text{m}$, respectively. The maximum power coupling coefficient is about 0.8 and the optimal fibre position relative to the waveguide is at (0.0, $3.6 \mu\text{m}$). The mode $1/e$ full width and depth are 8.8 and $5.6 \mu\text{m}$. Obviously, the mode profile in width is a good match to the fibre, and reducing the mode mismatch in depth is the key to increasing the coupling efficiency. The coupling might be improved by using a cylindrical lens [11].

Since the tolerance to the transverse (x or y) displacement of the fibre is much tighter than those to longitudinal and angular misalignments [6], we study only the influences of the horizontal and vertical offsets. The results are shown in Fig. 4. We see that 3 dB tolerances are about $4.8 \mu\text{m}$ and $4.0 \mu\text{m}$ for horizontal and vertical misalignments, respectively.

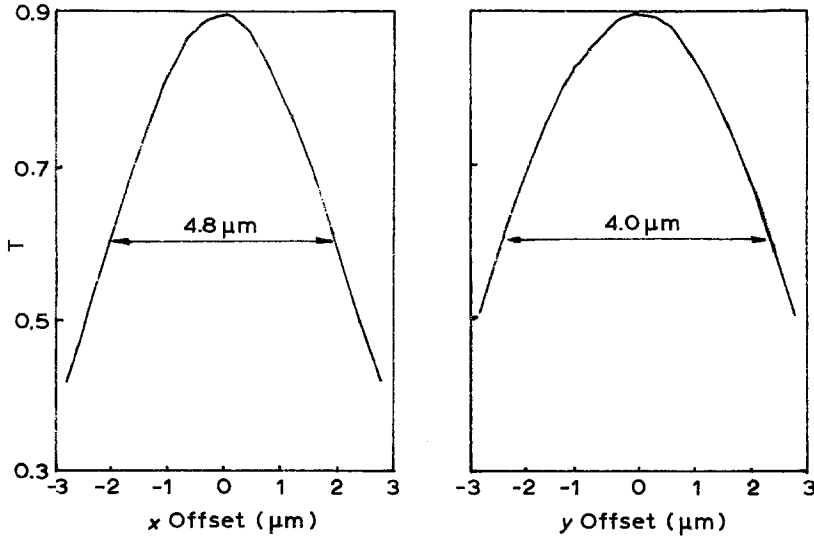


Figure 4 Calculated coupling coefficient versus transverse misalignments in the x and y directions.

4. Interaction or overlap integral between the optical mode and the applied signal

As an example, we consider a Z-cut LiNbO₃ intensity modulator with coplanar waveguide electrode, as shown in Fig. 5. Assuming that the electrode gap G is $6\mu\text{m}$, the centre conductor is $30\mu\text{m}$ wide, and the ground conductor width is over 1mm , and hence can be considered to be infinite, the y component of the applied electric field is given by [7]:

$$E_y = -(\epsilon_x/\epsilon_y)^{1/2} \psi_0 / K'(k) \operatorname{Re} \{ 2/S_1 / ([1 - (2Z/S_1)^2][1 - k^2(2Z/S_1)^2])^{1/2} \} \quad (8a)$$

where

$$K'(k) = K(1 - k^2)^{1/2} \quad (8b)$$

$$k = S_1 / (S_1 + 2G) \quad (8c)$$

$$Z = x + jy \quad (8d)$$

$K(k)$ is the elliptic integral of the first kind, ϵ_x and ϵ_y are the relative dielectric constants in the x and y directions, respectively, and ψ_0 is the applied voltage. The operation efficiency can be described by

$$\Gamma = \frac{G}{\psi_0} \iint E_y \phi^2 dS \quad (9)$$

The operation efficiency depends on the optical waveguide positions and the mode sizes.

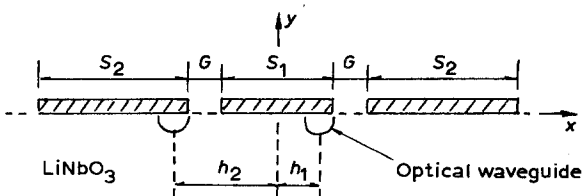


Figure 5 Cross-section of a Z-cut Mach-Zehnder amplitude modulator with coplanar waveguide electrode.

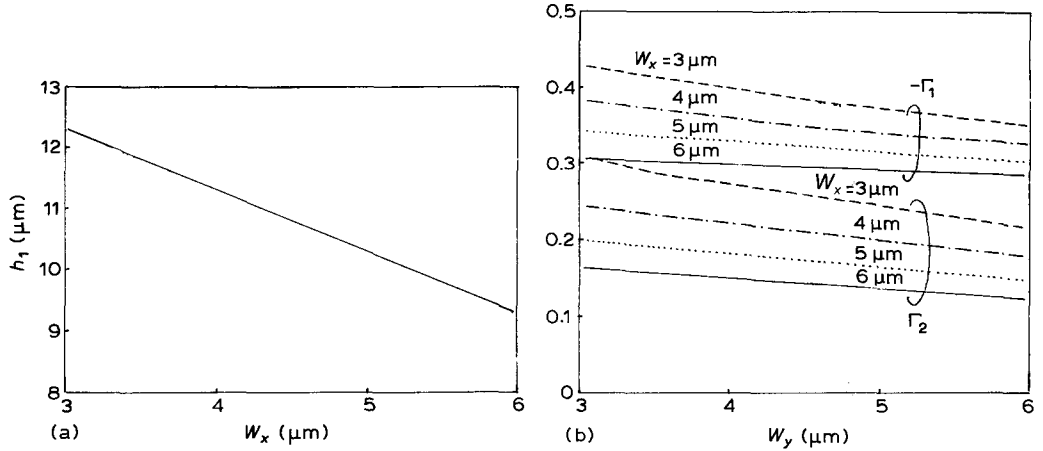


Figure 6 (a) Optimum waveguide position versus width parameter W_x ; (b) overlap integral in the case that the waveguides are at their optimal positions, where the subscripts stand for waveguide 1 and 2, respectively.

The optimal waveguide positions correspond to the largest value of Γ . According to our analysis, the optimal position of waveguide 2 (refer to Fig. 5) is at $-22.3 \mu\text{m}$ and does not change with the mode sizes. The optimal position of waveguide 1 is also independent of W_y , but is nearer the electrode gap for a narrow waveguide, as shown in Fig. 6a. Plotted in Fig. 6b is the overlap integral Γ as a function of the mode sizes when the optical waveguides are at their optimum positions. It can be seen that a relatively small mode is desirable to reduce the required drive voltage.

5. Discussion

Because of differences in fabrication details, measured diffusion parameters are quite different for the same fabrication conditions [12]; moreover, the available data reported in recent papers are inadequate to establish the relations between the diffusion parameters (diffusion width and depth, and index change) and the fabrication conditions (diffusion time, temperature, and titanium strip width and thickness). Although we cannot determine the expected set of fabrication conditions, the results reported in this paper can serve as a guide to designing low-loss and high-efficiency waveguide modulators.

The optimal position for a fibre coupled with optical waveguide could be found by using the method presented in Section 3. This allows proper choice of the groove dimensions of the silicon chip.

From Fig. 6b, one can see that the overlap integral continues to decrease with increased mode sizes, and optimum mode sizes do not exist. Most importantly, the optical waveguides should be placed near the centre line of the coplanar electrode but not exactly beneath the electrode edges. For a waveguide having the typical diffusion parameters discussed above, the optimum positions of the two waveguides are located at -22.3 and $10.9 \mu\text{m}$, respectively. Usually, the appropriate waveguide positions as illustrated in the literature [13–15] are at the electrode edges. For the electrode configuration discussed in this paper, $h_2 = S_1 + G + W/2 = 24 \mu\text{m}$, and $h_1 = S_1 - W/2 = 12 \mu\text{m}$. Therefore, there exist $1.7 \mu\text{m}$ and $1.1 \mu\text{m}$ differences for the two waveguides. If higher efficiency can be achieved by altering only the waveguide positions, it is worth doing.

According to our analysis, small mode sizes are desirable to reduce the required drive voltage, whereas the small mode sizes will result in poor optical coupling efficiency. Generally, for the small mode sizes the coupling can be improved by fabricating a lens on the fibre tip to transform the light beam from the fibre for a better match with the waveguide mode [16]. However, this technique is more difficult to implement because of the critical alignment tolerances. The coupling tolerances can be much relaxed by using a spherical lens and a single-mode optical fibre up-taper [17]. However, it should be noted that the choice of the mode sizes should satisfy two requirements. First, the waveguide must be a single-mode one. Additionally, it must provide a good guiding to achieve low propagation loss. In principle, using lensed fibre might be a viable approach to obtaining high coupling efficiency and low drive voltage.

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